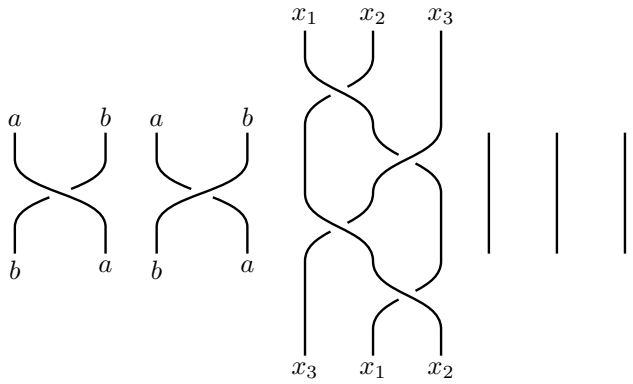
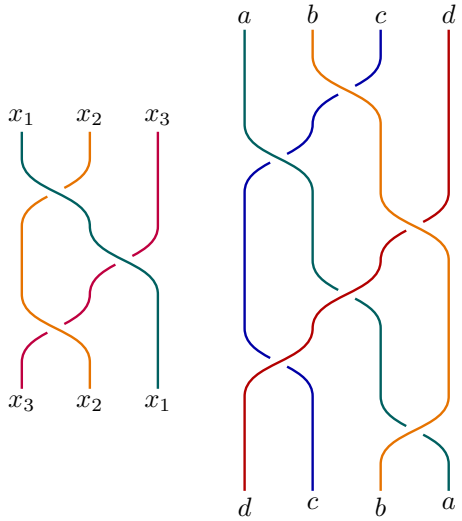
BraidTeX 0.5 — feature showcase

1. Crossings, identity, colours

Positive σ_1 (left strand over), negative σ_1^{-1}, a three-strand word, identities:

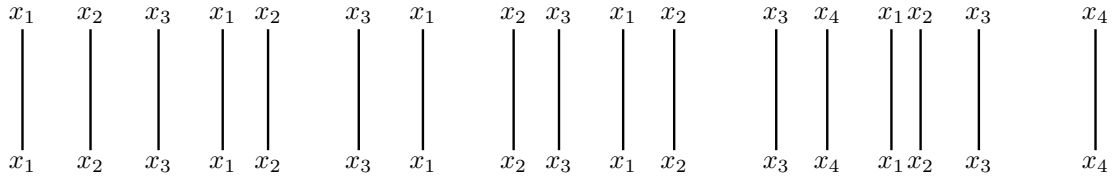


One colour per strand; colours follow strands; default palette:

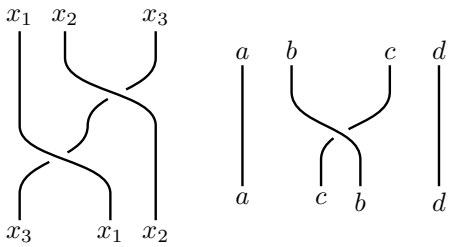


2. Automatic layout from parentheses

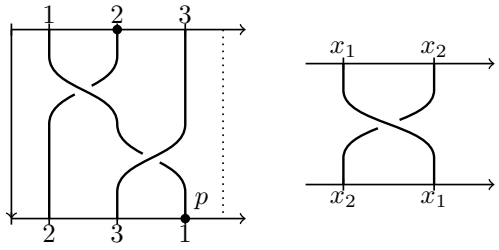
$x_1x_2x_3$, $(x_1x_2)x_3$, $x_1(x_2x_3)$, $(x_1x_2)(x_3x_4)$, $((x_1x_2)x_3)x_4$:



Different source and target parenthesizations:

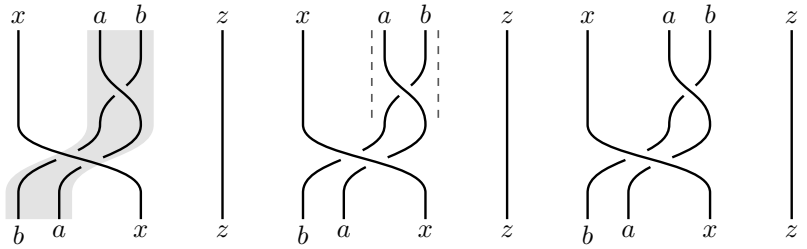


3. Axes and marked points

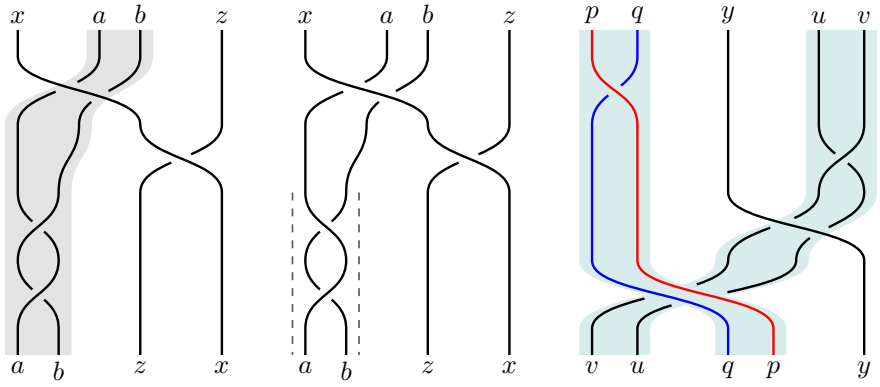


4. Composition: an inner braid replaces a strand

Outer braid σ_1 on $x\,y\,z$; strand y is replaced by the braid σ_1 on $a\,b$. **Mode 1, shadow** (default) shades the whole region the strand is replaced in; best when the ribbon travels through crossings with other strands. **Mode 2, separator** draws two dashed lines beside the inner braid only; best when the inner braid should read as a box inside a strand.

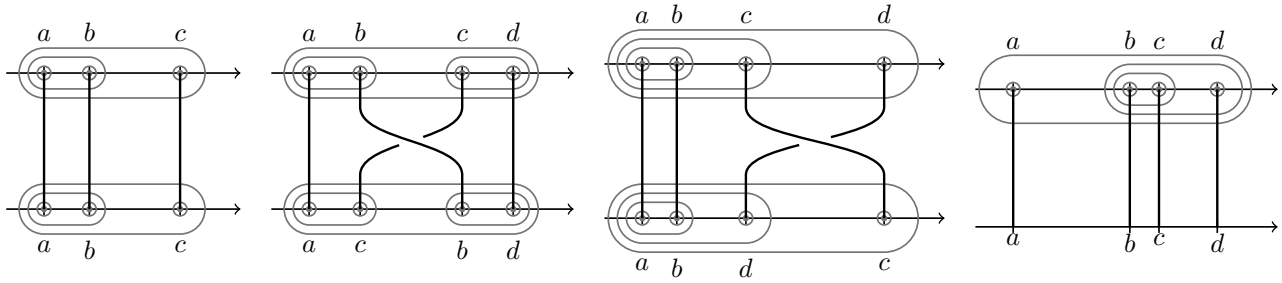


Inner braid at the bottom (`at=bottom`), shadow and separators; then two substitutions in one braid (colours, `shadowcolor`):



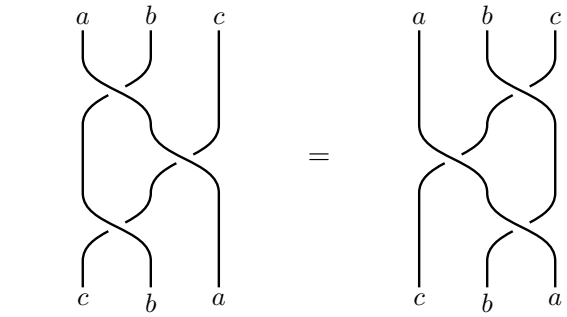
5. Little 2-disc boundary pictures

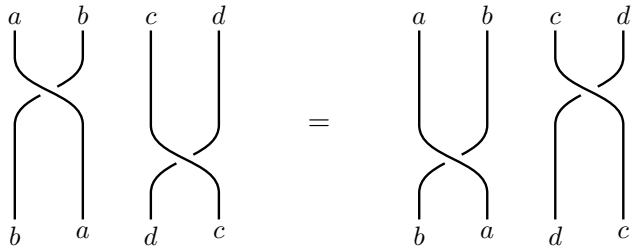
Each parenthesized group is one disc; $((ab)c)$ puts a, b in an inner disc, which together with c lies in the outer disc. Discs are drawn for the input (top) and the output (bottom) configuration.

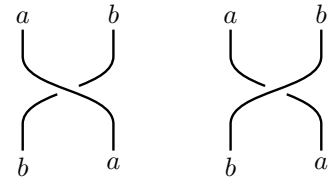


6. Standard identities

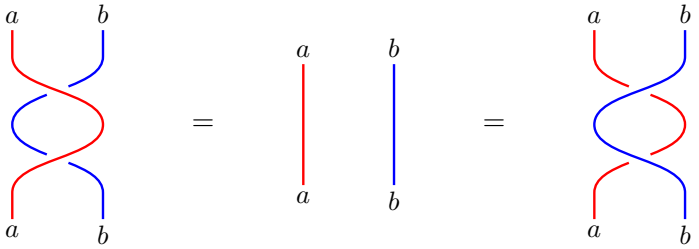
Yang–Baxter / braid relation, far commutativity, $\sigma\sigma^{-1} = 1$, the basic crossings:



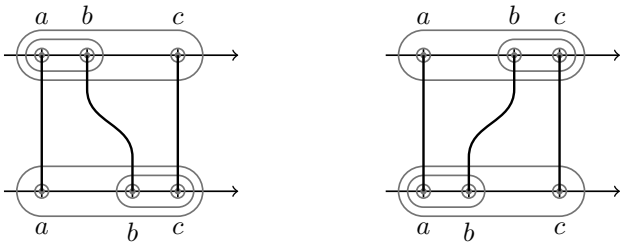




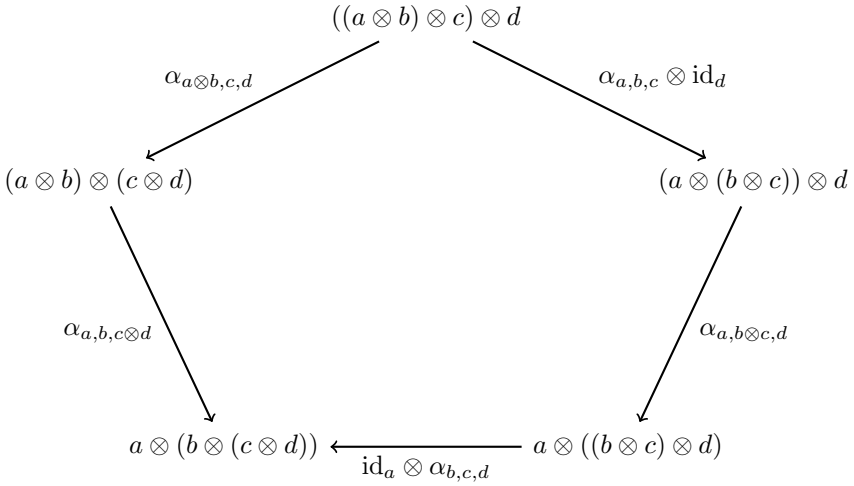
σ_1 $\qquad$ σ_1^{-1}



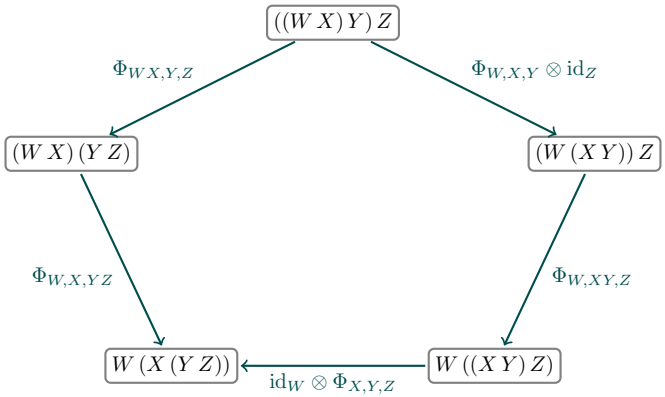
The associator as a braid between different parenthesizations (and its inverse):



Pentagon and hexagon (nodes, arrows and labels are generated; all configurable):



Restyled (own objects, no tensor sign, other associator symbol, node boxes, colour, scaled):



7. Inside expressions and inline

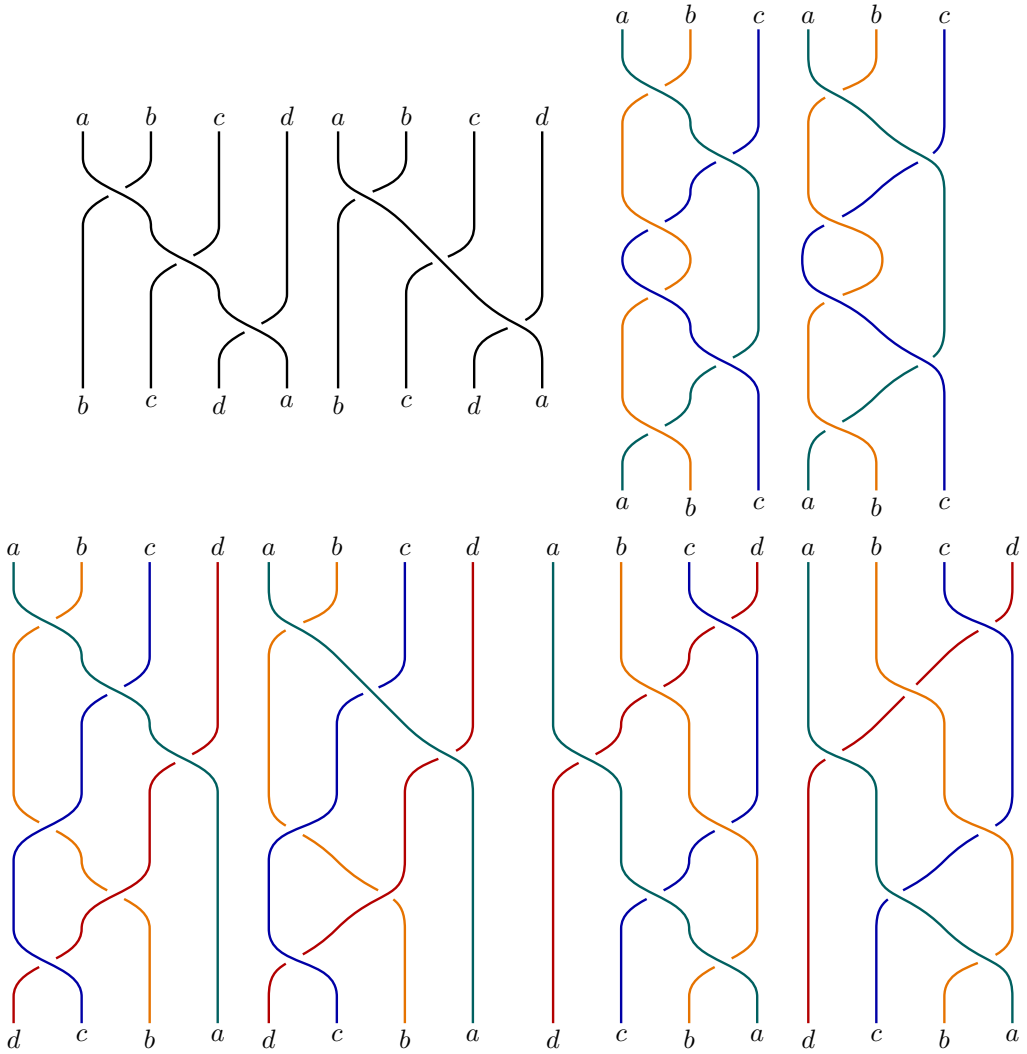
$$\psi(\tau) = \quad \cdot \quad , \tag{1}$$

$$\sigma_1 \quad = \tag{2}$$

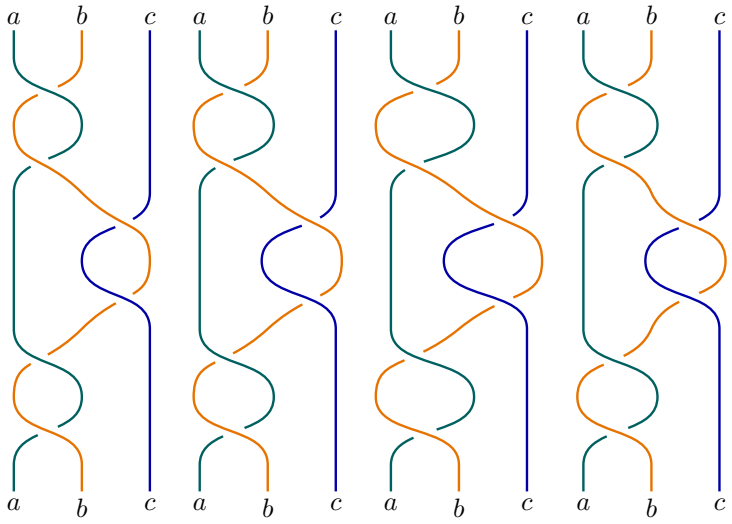
Inline: $\sigma_1 =$ and id $=$ sit on the math axis next to ordinary symbols.

8. Geometry modes: default snakes vs. natural strings

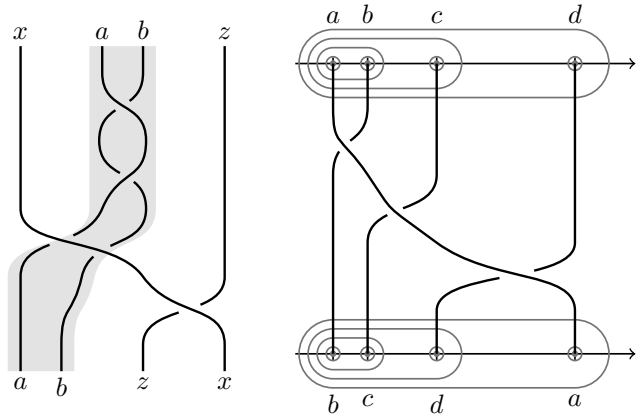
Same braids, `geometry=default` (left of each pair) and `geometry=natural` (right). In natural mode a strand with several crossings in a row is one smooth, nearly straight string; it only bends where it has to turn.



Back-and-forth twists: `wrap` sets how loosely a string winds around another (0, default, 0.2), and `tension=0.4` relaxes the straightening.

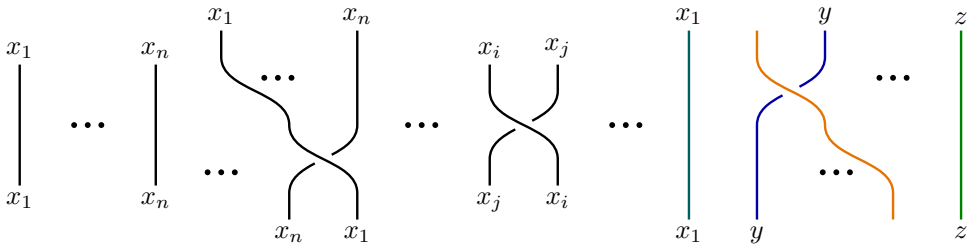


Natural mode combines with everything else (composition marks, discs, axes):

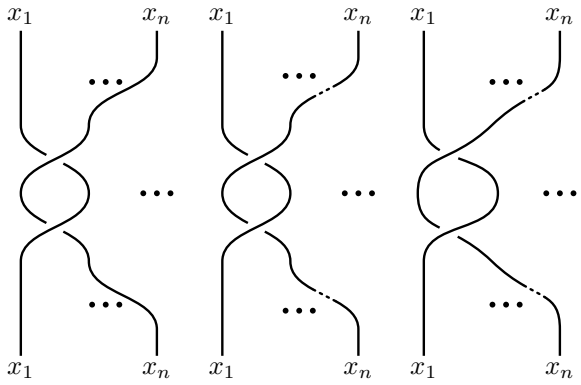


9. Large and infinite braids: gaps and unlabeled strands

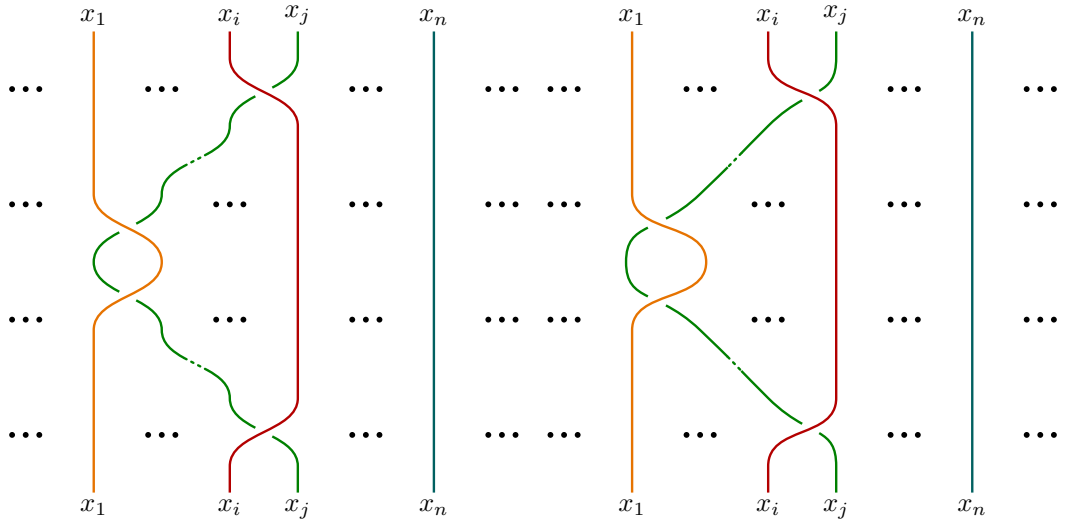
`...` marks omitted strands (a position like any other), `*` an unlabeled strand. Dots are aligned in rows shared by all gaps and keep clear of strands.



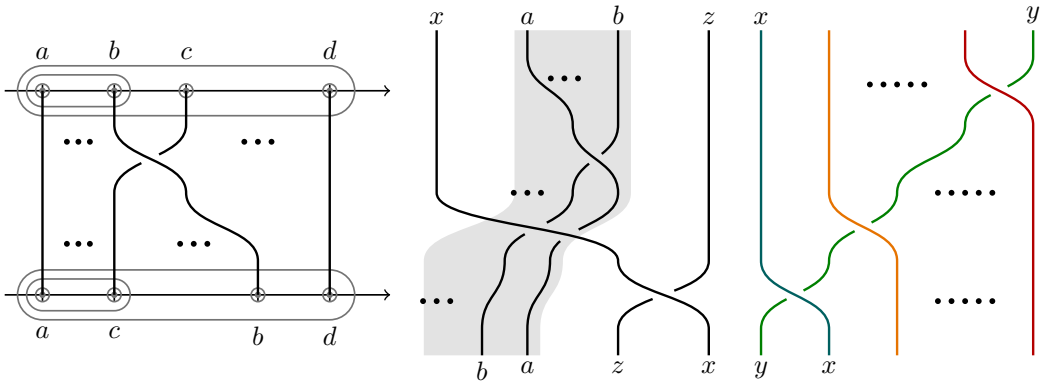
A strand winding around x_1 across the dots (solid, dotted, dotted with natural geometry):



The middle of a big braid, several gaps (aligned rows), two geometries:



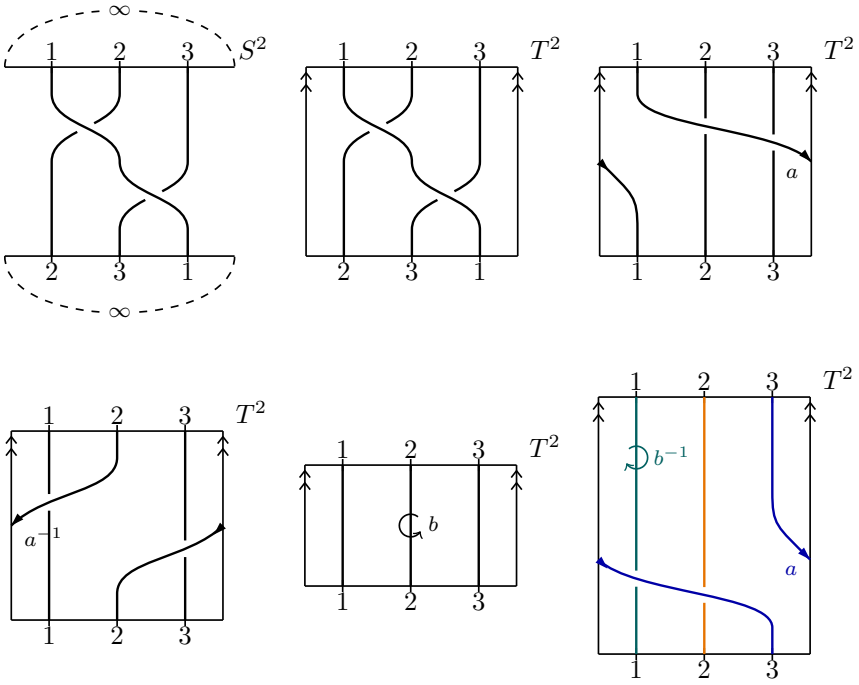
Gaps with discs, axes, a ribbon (shadow) and five dots:



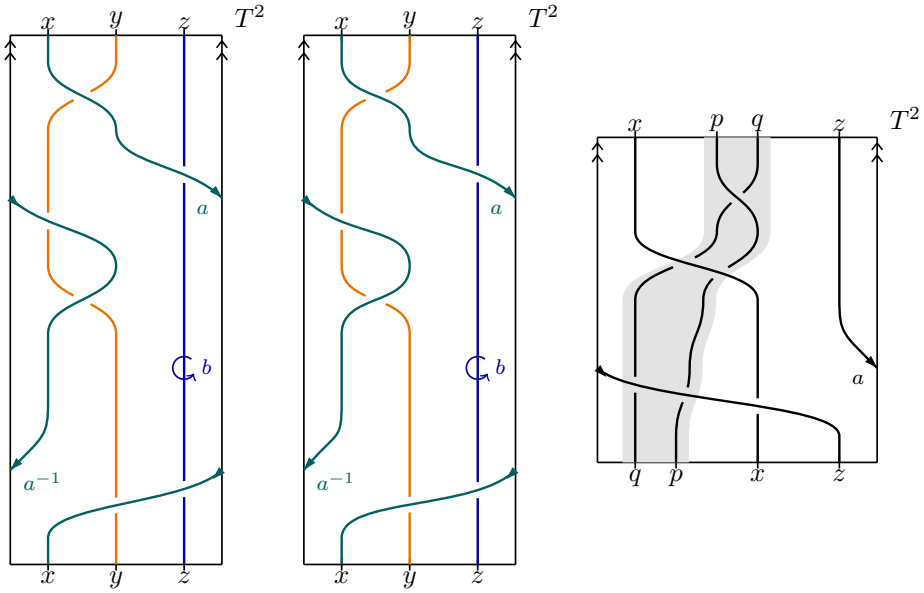
10. Braids on the sphere and the torus

`surface=sphere|torus` names the group. Sphere: the axes close through ∞ (no new generators, one new relation). Torus: identified edges (chevrons), loops `a`i (round the horizontal cycle, in front of

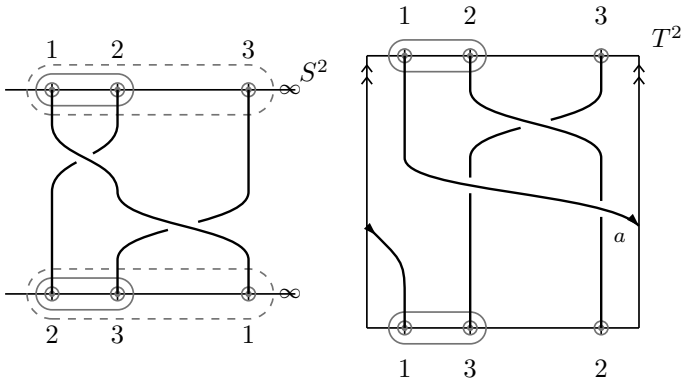
everything) and `bi` (round the depth cycle, a ring), inverses with a minus sign.



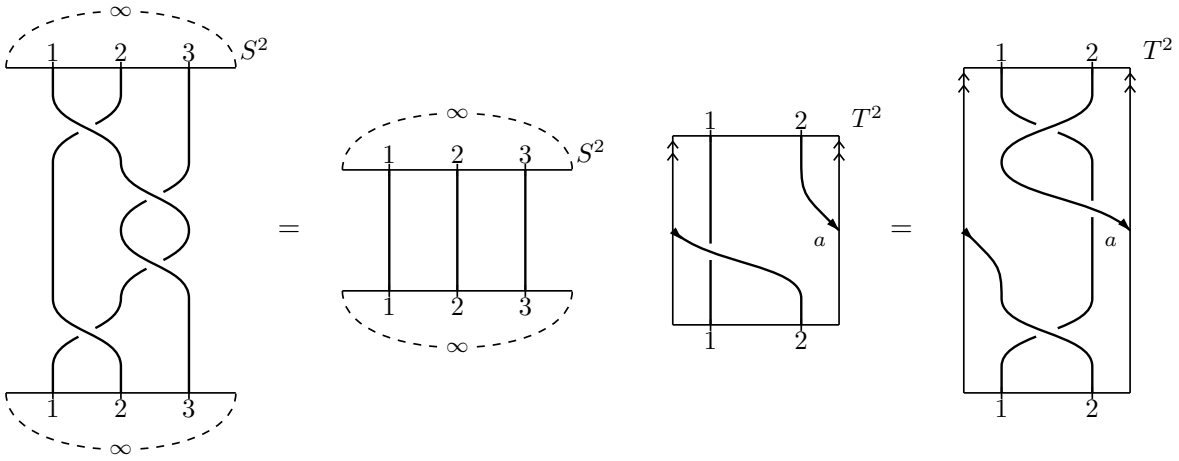
A longer torus word, both geometries, with colours, a gap and a ribbon (module insertion of a planar braid):

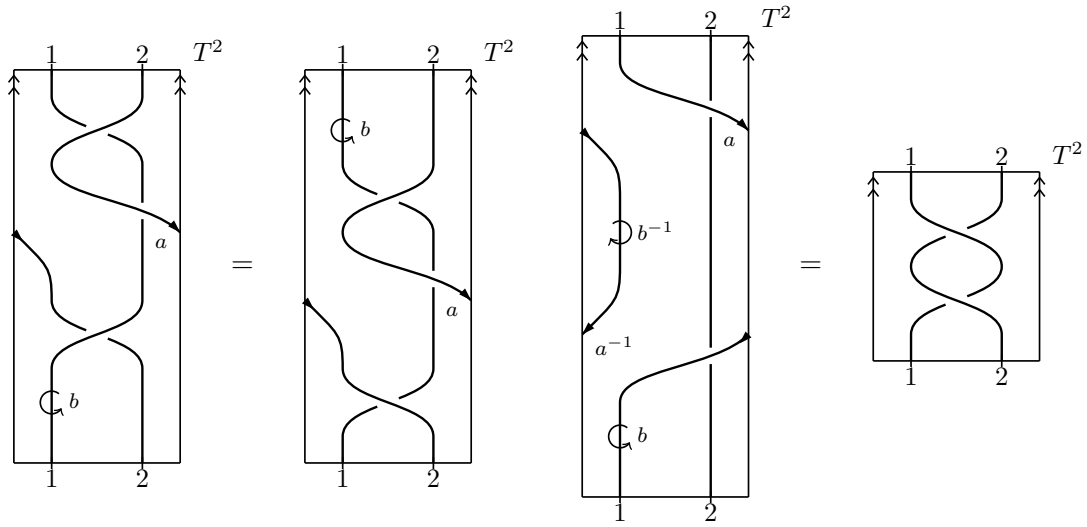


Little discs: dashed root disc with ∞ on the sphere, no root disc on the torus.



The relations, drawn from the verified words: the surface relation of the sphere, the conjugation formulas, the relation R4 and the torus relation TR.





11. Source scanning

```
\braid{\bin{x_1 x_2}\bcr{1}}   % shown literally, never transformed
```